$$\frac{cotg\,\delta}{tg\,\theta}=\frac{(\gamma+1)M_1^2-2(M_1^2 sen^2\theta-1)}{2(M_1^2 sen^2\theta-1)}$$

Função a ser zerada,

$$F=-2cotg\,\delta(M_1^2 sen^2\theta-1)+(\gamma+1)M_1^2 tg\,\theta$$
$$-2\,M_1^2 tg\,\theta\, sen^2\theta+2\,tg\,\theta=0$$

Derivada da função de teta,

$$\frac{d(tg\,\theta)}{d\theta}=\frac{d\left(\frac{sen\,\theta}{cos\,\theta}\right)}{d\theta}=sec^2\theta$$

$$\frac{d\,sen^2\theta}{d\theta}=2\,sen\,\theta\,cos\,\theta$$

$$tg\,\theta\,sen^2\theta=\frac{sen\,\theta}{cos\,\theta}sen^2\theta=(1-cos^2\theta)\frac{sen\,\theta}{cos\,\theta}=tg\,\theta-sen\,\theta\,cos\,\theta$$

$$\frac{d\,sen\,\theta\,cos\,\theta}{d\theta}=cos^2\theta-sen^2\theta=1-2\,sen^2\theta$$

$$\frac{dF}{d\theta}=-\frac{2}{tg\,\delta}M_1^2\,2\,sen\,\theta\,cos\,\theta+(\gamma+1)M_1^2 sec^2\theta$$
$$-2M_1^2(sec^2\theta-cos^2\theta+sen^2\theta)+2\,sec^2\theta$$

Mach 2 a direita da onda,

$$M_2^2=\frac{1}{sen^2(\theta-\delta)}\,\frac{1}{\frac{tg\,\theta}{tg(\theta-\delta)}\frac{(\gamma+1)}{2}-\frac{(\gamma-1)}{2}}$$